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Carbonation of fly ash in oxy-fuel CFB combustion

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Abstract

Oxy-fuel combustion of fossil fuel is one of the most promising methods to produce a stream of concentrated CO_2 ready for sequestration. Oxy-fuel FBC (fluidized bed combustion) can use limestone as a sorbent for in situ capture of sulphur dioxide. Limestone will not calcine to CaO under typical oxy-fuel circulating FBC (CFBC) operating temperatures because of the high CO_2 partial pressures. However, for some fuels, such as anthracites and petroleum cokes, the typical combustion temperature is above 900 °C. At CO_2 concentrations of 80–85% (typical of oxy-fuel CFBC conditions with flue gas recycle) limestone still calcines, but when the ash cools to the calcination temperature, carbonation of fly ash deposited on cool surfaces may occur. This phenomenon has the potential to cause fouling of the heat transfer surfaces in the back end of the boiler, and to create serious operational difficulties. In this study, fly ash generated in a utility CFBC boiler was carbonated in a thermogravimetric analyzer (TGA) under conditions expected in an oxy-fuel CFBC. The temperature range investigated was from 250 to 800 °C with CO_2 concentration set at 80% and H_2O concentrations at 0%, 8% and 15%, and the rate and the extent of the carbonation reaction were determined. Both temperature and H_2O concentrations played important roles in determining the reaction rate and extent of carbonation. The results also showed that, in different temperature ranges, the carbonation of fly ash displayed different characteristics: in the range 400 °C < $T \leqslant$ 800 °C, the higher the temperature the higher the CaO-to-carbonate conversion ratio. The presence of H_2O in the gas phase always resulted in higher CaO conversion ratio than that obtainable without H_2O. For $T \leqslant$ 400 °C, no fly ash carbonation occurred without the presence of H_2O in the gas phase. However, on water vapour addition, carbonation was observed, even at 250 °C. For $T \leqslant$ 300 °C, small amounts of $Ca(OH)_2$ were found in the final product alongside $CaCO_3$. Here, the carbonation mechanism is discussed and the apparent activation energy for the overall reaction determined.

Keywords: Oxy-fuel firing; CO_2; CFBC

1. Introduction

CO_2 accounts for about 50% of the anthropogenic greenhouse phenomenon in the earth's atmosphere. Huge quantities of CO_2 are produced from fossil fuel combustion every year. To reduce CO_2 emissions from combustion sources, one method is to burn fossil fuels in a mixture of oxygen and recycled flue gas, with sequestration of the CO_2 [1–6]. This approach eliminates almost all the atmospheric nitrogen, producing a flue gas that is composed primarily of CO_2, H_2O and small quantities of O_2, Ar, N_2, and trace gases like SO_2 and NO_x.

Use of recycled flue gas/oxygen mixtures for combustion in circulating fluidized bed combustion (CFBC) offers advantages over oxy-fuel combustion with conventional pulverized fuel (PF) firing technology. Oxy-fuel-fired PF units require high flue gas recirculation ratios to maintain acceptable furnace temperatures. CFBC can circulate large quantities of solids in the system. Oxy-fuel-fired CFBC takes advantage of this capability using external heat exchangers to remove heat from the system, significantly reducing the amount of recycled flue gas needed to maintain combustor temperatures at reasonable levels [7].

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The reduced flue gas recirculation ratio should result in significant Boiler Island cost savings. This unique feature of fluidized bed combustors means that much higher percentages of oxygen can be used in the recycled flue gas/oxygen mixtures than would be possible in alternate oxy-fuel firing applications. This allows the design and construction of more compact, relatively less expensive CFB boilers. Economic analysis indicates that the proposed oxygen-firing technology in circulating fluidized bed boilers could be developed and deployed economically in the near future [8].

2. Carbonation of fly ash under oxy-fuel FBC combustion conditions

A major benefit of fluidized bed combustion is the ability to capture SO_2 in situ by adding limestone to the combustor. The sulphur capture occurs in two steps, first calcination and then sulphation of Ca-based sorbents:

$$CaCO_3 + \text{heat} \rightarrow CaO + CO_2 \tag{1}$$

$$CaO + SO_2 + 1/2O_2 \rightarrow CaSO_4 + \text{heat} \tag{2}$$

The calcination temperature of $CaCO_3$ depends on the CO_2 partial pressure in the surrounding gas. For air firing, the CO_2 content of the flue gas is under 20%, and limestone calcines at ~700 °C, which is well below the typical CFB operating temperature of 800–850 °C. With oxygen firing, however, the CO_2 content is above 70% leading to a calcination temperature of above 870 °C [8].

In oxy-fuel CFB combustion, because of higher local O_2 concentration, it is possible that combustor temperature can rise to close to 900 °C locally. Also, fuels such as anthracites and petroleum cokes are typically combusted at above 870 °C in CFB combustors. Therefore, calcination of limestone sorbent will occur. The typical Ca utilization ratio in CFBCs is about 30–45%. Therefore, significant quantities of CaO may exist in both CFBC bed ash and fly ash. If the ash is cooled to below the calcination temperature while exposed to the high-CO_2 environment, carbonation of the unused CaO may occur:

$$CaO + CO_2 \rightarrow CaCO_3 \tag{3}$$

In a conventional CFBC, the flue gas leaving the cyclone is cooled in the convective pass followed by an air preheater. In the convective pass, fly ash typically deposits on the tube banks. With oxy-fuel CFBC firing, the CO_2 content is high (>80%), so carbonation can occur over a broader temperature range (~760 °C for oxy-fuel CFB combustion) and at a much higher rate. If carbonation of CaO in the ash deposits has occurred, the hardness and strength of the deposits can significantly increase, making the ash deposits on the heat transfer surfaces more difficult to remove. It is the aim of this paper to study the factors that influence the carbonation process under oxy-fuel conditions.

When fly ash travels through the convective pass of the boiler, ash particles can deposit on heat transfer surfaces. The rate of deposit build-up depends on a number of factors, such as flue gas velocity, temperature, etc. Such deposits will decrease convective and radiative heat transfer coefficients. In many cases, even a small amount of ash deposited on a surface decreases local heat transfer rates by factors of three or more [9].

Apart from their impact on heat transfer, ash deposits in boilers may cause operational problems and boiler maintenance issues, including plugging, corrosion and structural damage.

Numerous studies have been carried out to investigate the influence of deposits in boilers [9–15]. One study [10] looked at the ash deposition in a pulverized coal-fired power plant after high-calcium lignite combustion, and found that the following calcium mineral phases were dominant in deposits: anhydrite [$CaSO_4$], calcite [$CaCO_3$], portlandite [$Ca(OH)_2$], and lime [CaO]. Russell et al. [11] studied the role of CaO on the formation of ash and deposits in PF combustion. This study showed that the nature of the calcium mineral was very important for the formation of ash deposits. In addition, carbonation has been identified as a cause of fouling in CFBC boilers firing high levels of petroleum coke [16–18]. CaO content in CFBC fly ashes is generally high (~20% or more). Under oxy-fuel CFB combustion conditions, CO_2 concentration in the flue gas is high, and it is easier for carbonation to occur in oxy-CFB combustion than in conventional air-fired CFBCs. This will have impact on the ash deposits in the boiler.

A number of investigations [19–24] exist on carbonation of limestone-derived materials, but the main focus of these studies is on capturing CO_2 from combustion and/or gasification processes, since the use of carbonation/calcination cycles of $CaO/CaCO_3$ is emerging as a viable alternative technique to capture CO_2 generated in the combustion of fossil fuels for power generation. Specifically, the choice of natural limestones as CO_2 carriers is an attractive option because they are cheap and abundant materials.

There are a few investigations on the carbonation of fly ash in oxy-fuel combustion. ALSTOM Power [8] has done some work on CFBC ash carbonation in oxy-fuel combustion, but the possible effect of H_2O vapour on carbonation was not considered. Recently, Chen et al. [25] studied calcination and sintering characteristics of limestone-derived material under O_2/CO_2 combustion conditions. They found that pore volume and specific surface area of the limestone calcined under O_2/CO_2 combustion conditions were lower than for calcines obtained under air-firing conditions (~25%). What this means for the carbonation of the fly ash needs to be studied further. Previous studies [26,27] have shown that during calcination, limestone particles generally do not shrink. In consequence, typical non-porous limestone particles will become highly porous (~36%) under air-fired atmospheric FBC conditions (800–950 °C).

Since actual flue gas produced in fossil fuel-fired power plants inevitably contains H_2O, its influence on ash carbonation must be evaluated. In oxy-fuel FBC combustion flue gas, H_2O may vary from 5–15%, and small amounts of O_2

and N_2 and trace amounts of CO, SO_x, NO_x, etc., other than CO_2, may also be present. In this work, carbonation of fly ash with simulated flue gas containing CO_2, H_2O vapour, O_2 and N_2 was conducted under oxy-fuel CFB combustion conditions.

3. Experimental

A fly ash sample collected from the baghouse of a CFB boiler from a Canadian power plant was carbonated in a thermogravimetric analyzer (TGA) using a Cahn TGA. The composition of the fly ash is given in Table 1. The ash was generated when firing a mixture of petroleum coke and coal (60/40) with limestone added as sulphur sorbent and, therefore, is mostly sorbent derived. The ash is fine, with particles smaller than 75 μm accounting for about 95% of the total weight. Since char carbon contained in the fly ash may interfere with the measurement of sample weight in the TGA, the ash was heated in an oven at 800 °C for 2 h to burn out the char carbon content. Free lime content of the fly ash was determined by a modified sucrose method based on ASTM C-25. Free lime is defined as the sum of CaO and $Ca(OH)_2$ expressed as CaO. Three determinations were made for this ash and the result was 50.09% ± 0.2%.

All experiments followed the same procedure. A known amount of fly ash sample (between 20 and 30 mg) was loaded into the TGA in a shallow platinum sample pan (12.7 mm diameter, 2 mm deep). The TGA was flushed with pure N_2 at 300 mL/min for 15 min before heating began. The fly ash sample was then heated to the desired temperature under N_2 environment. Once the preset temperature was reached, the TGA gas supply was switched to synthetic flue gas containing CO_2, O_2, H_2O and N_2. The flow rate was maintained at 300 mL/min throughout the test. Sample weight and temperature were monitored and recorded continuously. Table 2 gives the experimental conditions.

Table 2
Experimental conditions

Temperature (°C)	250–800
CO_2 concentration (%)	80
O_2 concentration (%)	1
H_2O vapour concentration (%)	0–15
N_2, %	4–19

4. Results and discussion

4.1. 400 °C < T ⩽ 800 °C

Fig. 1 shows carbonation results obtained for a temperature range of 400–800 °C. Several observations can be made for this temperature range. First, H_2O was shown to play an important role in the carbonation of CaO. The carbonation conversion ratio of CaO with H_2O vapour present was always higher than that without H_2O at any given temperature, with the difference more pronounced at lower temperatures. For example, when the carbonation reaction approached its end, the difference in absolute CaO conversion ratio at 500 °C was about 47% higher with 15% H_2O vapour in the synthetic flue gas than that obtained without H_2O vapour present. At 800 °C the difference was <10%.

Second, temperature has a strong effect on the rate of the carbonation reaction. Higher temperatures correspond to increased carbonation reaction rate. The carbonation reaction rate was highest at the start of the reaction (time 0). As the reaction proceeded, the rate decreased continuously, because of inhibition by the product layer (transport resistance) and also diminishing availability of CaO. Comparing the two curves in Fig. 1a and b at 800 and 700 °C with H_2O at 15%, it took about 3 min at 800 °C and 9 min at 700 °C, for the rate to obviously slow down. This confirmed that higher temperatures led to higher carbonation rates.

Third, the difference in CaO conversion ratio and carbonation rate between 15% H_2O and 8% H_2O was not significant, as shown in Fig. 1a and c, although the conversion ratio and carbonation rate at 15% H_2O were slightly higher than at 8% H_2O. This showed the weak effect of H_2O vapour concentration, at least in the range of 8–15% as covered in this work.

Fourth, carbonation was fast at the start, and then the rate gradually reduced as the reaction proceeded. This was true for carbonation reactions at all temperature ranges covered in this work. This phenomenon was observed under all test conditions. This is attributed to the formation of $CaCO_3$ product layer that inhibits direct contact between gaseous CO_2 and lime. A very similar phenomenon has been observed in CO_2 carbonation work for looping cycles, i.e. there is a fast carbonation period lasting minutes and then a much slower one which continues for long periods [28].

After the carbonation tests the fly ash samples were analyzed. It was found that the $CaCO_3$ was the only product when carbonation temperature was higher than 400 °C.

4.2. 300 °C < T ⩽ 400 °C

Carbonation of CaO at 400 and 350 °C is shown in Fig. 2a and b. Some of the characteristics in Fig. 2 are

Table 1
Fly ash analysis (wt%)

Compound	SiO_2	Al_2O_3	Fe_2O_3	TiO_2	P_2O_5	CaO	MgO	SO_3	Na_2O	K_2O	Loss on fusion
Wt%	2.56	1.35	0.67	0.08	0.03	55.44	1.04	11.85	0.2	0.12	26.58

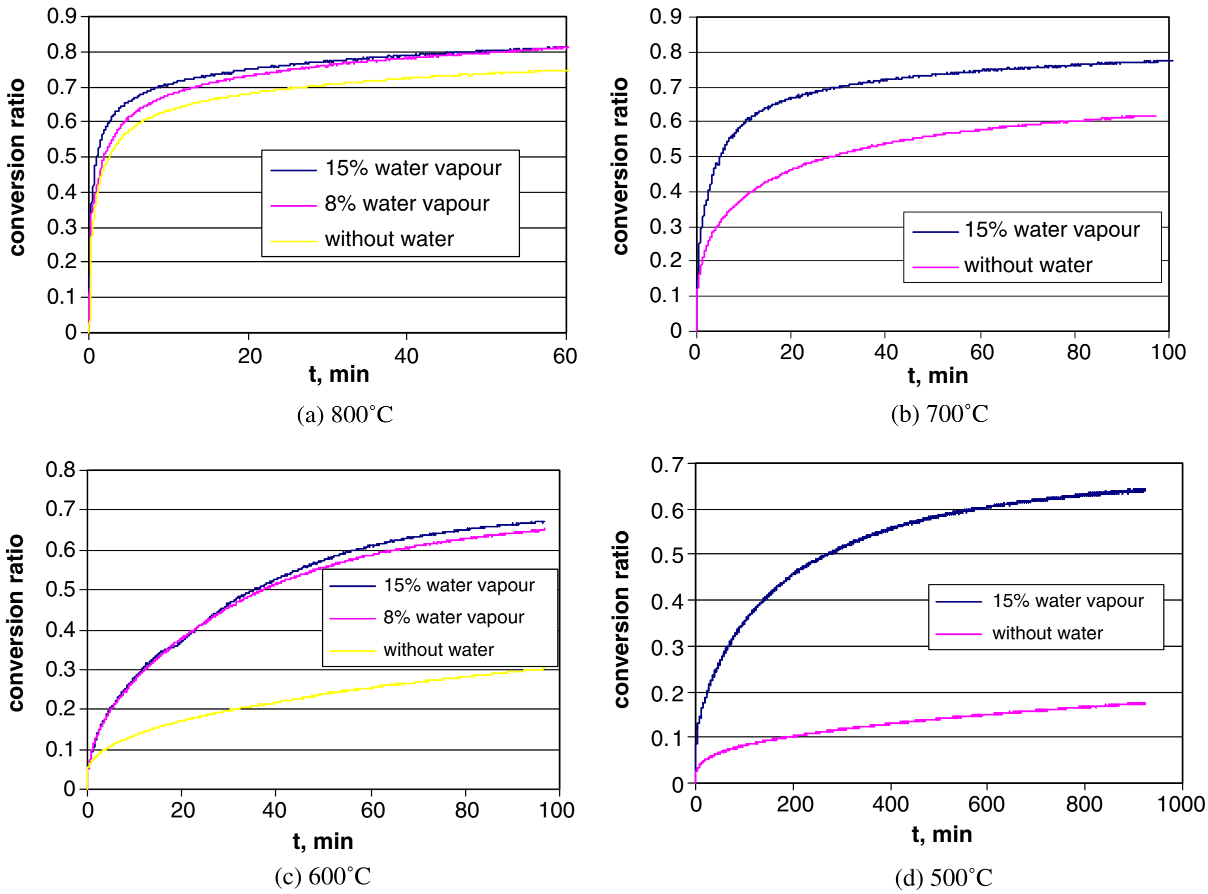

Fig. 1. Carbonation of fly ash: (a) 800 °C, (b) 700 °C, (c) 600 °C, (d) 500 °C.

similar to those seen in Fig. 1. Thus, with higher temperatures, the carbonation reaction rate was greater and the extent of CaO conversion ratio was also higher for any given time. For 15% H_2O, the conversion ratio at 16 h was about 0.4 at 400 °C and less than 0.3 at 350 °C, a difference of > 10% for a 50 °C decrease. Alternatively, at 20% CaO conversion, the time required was about 78 min at 400 °C compared to about 358 min required to reach the same conversion level at 350 °C with 15% H_2O in the synthetic flue gas.

Compared to results in Fig. 1, Fig. 2 showed some unique characteristics. First, when $T \leqslant 400$ °C, TGA testing showed that there was no noticeable carbonation reaction for 1000 min without H_2O, even when the CO_2 concentration was as high as 80%. However, carbonation readily occurred when H_2O (8% or 15%) vapour was added to the synthetic flue gas. Since the reaction $CaO + CO_2 = CaCO_3$ does not take place directly in this temperature range, other mechanisms must exist to account for the carbonation observed in Fig. 2 [29,30]. Second, higher H_2O concentration led to higher CaO conversion ratio and carbonation rate, the same trend as shown in Fig. 1a and c for higher temperatures. However, the difference in conversion ratio and carbonation rate between 15% H_2O and 8% H_2O became more pronounced for the lower temperature range (300–400 °C). This demonstrated that H_2O concentration plays an increasingly more important role as temperature decreases.

When the fly ash samples were analyzed after the carbonation test, the final product was only $CaCO_3$, similar to samples carbonated at higher temperatures described above, with no evidence of $Ca(OH)_2$ formation.

4.3. 250°C $\leqslant T \leqslant$ 300°C

The conversion ratio and carbonation rate decreased rapidly in this temperature range, Fig. 3, when compared with carbonation tests conducted at higher temperatures (Figs. 1 and 2). For example, for 15% H_2O, the conversion ratio was only about 0.21 at 300 °C and less than 0.09 at 250 °C after 1000 min. It was evident that temperature played a more important role in the carbonation of CaO in the fly ash. When carbonation temperature was >300 °C, the final product was $CaCO_3$. But in this temperature range (250 °C $\leqslant T \leqslant$ 300 °C), both $CaCO_3$ and $Ca(OH)_2$ were found in the treated samples. When both $CaCO_3$ and $Ca(OH)_2$ were present in the final ash sample, the molar ratio of $CaCO_3$/$Ca(OH)_2$ could be determined. This ratio is shown in Table 3. It was found that:

- at the same temperature, the quantity of $Ca(OH)_2$ increased with the increase in concentration of H_2O and,

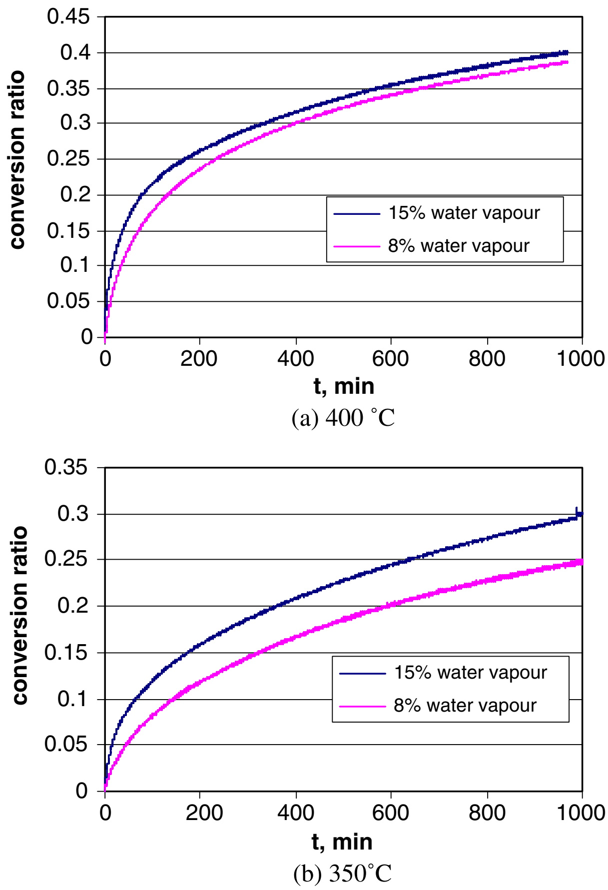

Fig. 2. Carbonation of fly ash: (a) 400 °C, (b) 350 °C.

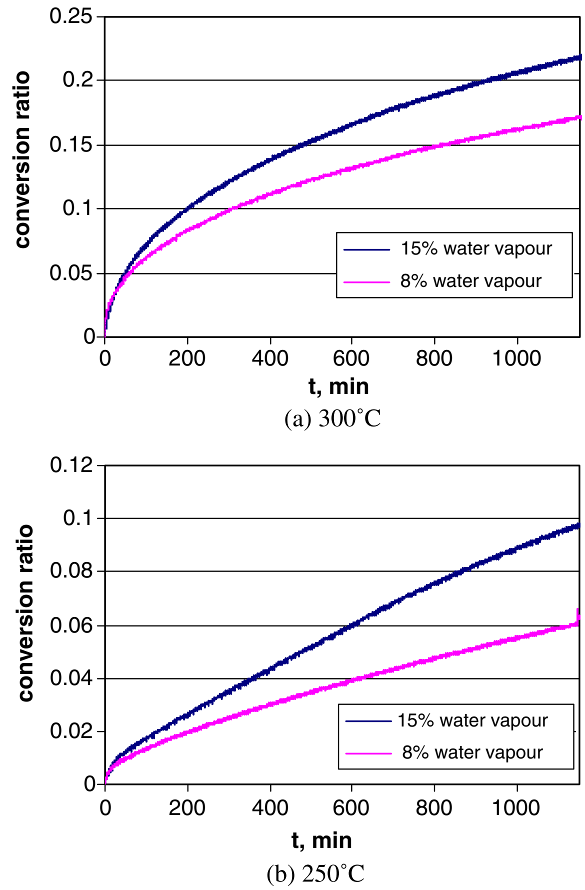

Fig. 3. Carbonation of fly ash: (a) 300 °C, (b) 250 °C.

- at the same concentration of H_2O, the quantity of $Ca(OH)_2$ decreased with increasing temperature.

5. Carbonation mechanism

As shown in Figs. 1–3, carbonation of CaO with and without H_2O vapour present in the synthetic flue gas showed significant differences. Most noticeably, carbonation of CaO did not occur when $T \leqslant 400°C$ without H_2O, but at 15% and 8% H_2O present, significant CaO conversion to $CaCO_3$ was achieved. It appears that carbonation of CaO in real coal-fired flue gas proceeds *via* reaction (3) and is catalyzed by H_2O vapour, presumably by the formation of a transient hydroxide species. Here we suggest that when water vapour is present in the flue gas, the actual carbonation process of CaO occurs via the following two-step reaction:

First, CaO reacts with H_2O to form $Ca(OH)_2$:

$$CaO + H_2O = Ca(OH)_2 \tag{4}$$

$Ca(OH)_2$ then reacts with CO_2:

$$Ca(OH)_2 + CO_2 = CaCO_3 + H_2O \tag{5}$$

Fig. 4 shows that, at H_2O partial pressure less than 20 kPa and temperature >400 °C, $Ca(OH)_2$ cannot exist as a stable compound. However, it can be assumed that $Ca(OH)_2$ will still form when H_2O encounters CaO as a transient intermediate. The concentration of $Ca(OH)_2$ intermediate could be very low, since $Ca(OH)_2$ calcines to CaO quickly. The higher the temperature is, the shorter the existence of

Table 3
Molar ratio of $CaCO_3/Ca(OH)_2$ at different conditions

Conditions	250 °C, 15% H_2O	250 °C, 8% H_2O	300 °C, 15% H_2O	300 °C, 8% H_2O
$CaCO_3/Ca(OH)_2$	0.241	0.457	2.394	5.400

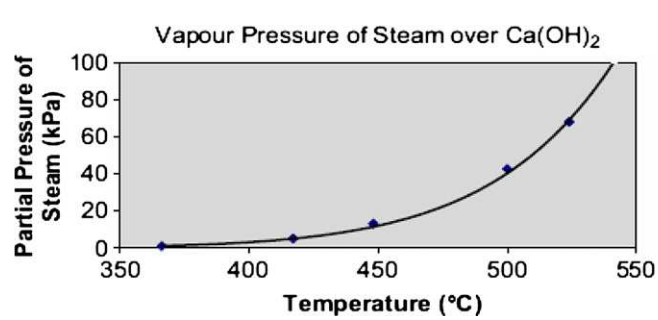

Fig. 4. Partial pressure of water vapour over $Ca(OH)_2$.

$Ca(OH)_2$ transient species will be, and the less important its contribution to the carbonation process will be. However short the existence of $Ca(OH)_2$ is, if a CO_2 molecule meets $Ca(OH)_2$, $CaCO_3$ should result according to reaction (5). As noted elsewhere [29,30], carbonation of $Ca(OH)_2$ is much faster than carbonation of CaO. Therefore, the presence of H_2O vapour in the flue gas increases the carbonation rate of CaO contained in the fly ash as observed with results in this study.

It should be noted that the contribution of reaction (3) becomes less important when temperature decreases. This can be seen in Fig. 1 which showed that lower temperatures led to more pronounced differences between tests done with and without H_2O. For example, when the carbonation reaction tended to stabilize, there was <10% carbonation ratio difference between 15% H_2O and 15% N_2 at 800 °C, but at 500 °C the difference was 47%.

At $T \leqslant 400$ °C, there was no detectable carbonation reaction of CaO without H_2O. But when 8–15% of H_2O vapour was added to the synthetic flue gas, carbonation of CaO was obvious. This observation supports the proposed second mechanism that the carbonation of CaO when H_2O vapour was present also includes the two-step route: $CaO \rightarrow Ca(OH)_2 \rightarrow CaCO_3$. From experimental results shown in Fig. 3, one might conclude that reaction (4) was faster than (5), since the difference in final CaO reaction products existed, i.e., there was no $Ca(OH)_2$ in the carbonated fly ash samples when $T > 300$ °C, but some $Ca(OH)_2$ remained when $T \leqslant 300$ °C, and the amount of $Ca(OH)_2$ increased as the temperature decreased.

6. Activation energy

For carbonation of the ash, the activation energy can be calculated by the equation [31]:

$$\ln\frac{t_1}{t_2} = \frac{E}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right) \tag{6}$$

where t_1, t_2, are reaction time; T_1, T_2, are reaction temperatures in K; E is activation energy in kJ/mol; and R is the gas constant, 8.31 kJ/mol K. The degree of free CaO conversion used in the estimation of E is in the range of 5% to 30%.

Table 4 gives the estimated activation energy. It can be seen that the effective activation energy of carbonation for the ash used was smaller when H_2O was present than when it was not. Apparently, H_2O vapour acted as a catalyst for the carbonation of CaO. This is consistent with experimental observations in terms of CaO to $CaCO_3$ conversion ratio, which are shown in Figs. 1–3. It is clear that the carbonation conversion ratio and reaction rate of the CaO contained in the fly ash were enhanced by the presence of H_2O.

Table 4
Activation energy for carbonation of baghouse ash

Conditions	Activation energy, E (kJ/mol)
15% H_2O	98.3
No H_2O	180.4

7. Conclusions

A series of tests were conducted in a TGA to study the potential for fly ash carbonation under oxy-fuel combustion conditions. The results indicated that H_2O played an important role in the carbonation of CaO. When H_2O was present in the reaction gas, the carbonation conversion ratio of CaO was higher than that obtained without H_2O at any given time under typical oxy-fuel CFBC operating conditions. The relative importance of H_2O increased as temperature decreased. The results showed that no noticeable carbonation of CaO occurred without H_2O when $T \leqslant 400$ °C. When the synthetic flue gas had 8–15% of H_2O, carbonation of CaO occurred at even lower temperatures (~250 °C). Temperature was another important factor that determined the carbonation rate and conversion ratio *vs.* time. Higher temperatures increased both extent of CaO conversion and carbonation reaction rate.

When $T > 300$ °C the final product was $CaCO_3$ and below this temperature some $Ca(OH)_2$ remained in the carbonated fly ash samples. Based on the above findings, it was proposed that, besides reaction (3), the carbonation process of CaO may also include a two-step route when H_2O vapour is present: $CaO \rightarrow Ca(OH)_2 \rightarrow CaCO_3$. The activation energy was calculated to be 98.3 kJ/mol in the presence of 15% H_2O and 180.4 kJ/mol without H_2O.

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